parameters.bsn

Purpose

`parameters.bsn` holds the basin-wide numeric coefficients (initial soil water, surface runoff lag, nutrient cycling rates, channel routing factors, lapse rates, CO2, and so on). Values are read into `bsn_prm` (`type basin_parms`) and consulted by hydrology, nutrient, erosion, and routing routines.

Source

- Reader: `src/basin_read_prm.f90`
- Type definition: `src/basin_module.f90`, `type basin_parms`

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Line 3: 44 values, list-directed, read into `bsn_prm` in declaration order. The last two values are integers; the rest are reals.

#	Field	Type	Default	Units	Description
1	evlai	real	3.0	none	leaf area index above which no soil evaporation occurs
2	ffcb	real	0.0	fraction	initial soil water content as a fraction of field capacity
3	surlag	real	4.0	days	surface runoff lag time
4	adj_pkr	real	1.0	none	peak rate adjustment factor in the subbasin
5	prf	real	484.0	none	peak rate factor for the peak rate equation
6	spcon	real	0.0	none	not used
7	spexp	real	0.0	none	not used
8	cmn	real	0.003	1/day	rate factor for mineralization of active organic N
9	n_updis	real	20.0	none	nitrogen uptake distribution parameter
10	p_updis	real	20.0	none	phosphorus uptake distribution parameter
11	nperco	real	0.10	0..1	nitrate percolation coefficient
12	pperco	real	10.0	0..1	phosphorus percolation coefficient
13	phoskd	real	175.0	none	phosphorus soil partitioning coefficient
14	psp	real	0.40	none	phosphorus availability index
15	rsdco	real	0.05	none	residue decomposition coefficient
16	percop	real	0.5	0..1	pesticide percolation coefficient
17	msk_co1	real	0.75	none	Muskingum storage time constant for bankfull depth
18	msk_co2	real	0.25	none	Muskingum storage time constant for low flow
19	msk_x	real	0.20	none	Muskingum weighting factor for inflow vs outflow
20	nperco_lchtile	real	0.5	none	N concentration coefficient for tile flow and bottom-layer leaching
21	evrch	real	0.60	none	reach evaporation adjustment factor
22	scoef	real	1.0	0..1	channel storage coefficient
23	cdn	real	1.40	none	denitrification exponential rate coefficient
24	sdnco	real	1.30	none	denitrification threshold (fraction of field capacity)
25	bact_swf	real	0.15	fraction	fraction of manure containing active colony-forming units
26	tb_adj	real	0.0	none	adjustment factor for sub-daily unit hydrograph base time
27	cn_froz	real	0.000862	none	parameter for frozen-soil adjustment on infiltration/runoff
28	dorm_hr	real	-1.0	hours	dormancy time threshold
29	plaps	real	0.0	mm/km	precipitation lapse rate per km of elevation difference
30	tlaps	real	6.5	deg C/km	temperature lapse rate per km of elevation difference
31	nfixmx	real	20.0	kg/ha/day	maximum daily nitrogen fixation
32	decr_min	real	0.01	1/day	minimum daily residue decay
33	rsd_covco	real	0.75	none	residue cover factor for computing cover fraction
34	urb_init_abst	real	1.0	none	maximum initial abstraction for urban areas (Green-Ampt)

#	Field	Type	Default	Units	Description
35	petco_pmpt	real	100.0	percent	PET adjustment for Penman-Monteith and Priestley-Taylor
36	uhalpha	real	1.0	none	alpha coefficient for the gamma-function unit hydrograph
37	eros_spl	real	0.0	none	coefficient of splash erosion (typical range 0.9 to 3.1)
38	rill_mult	real	0.0	none	rill erosion coefficient
39	eros_expo	real	0.0	none	exponential coefficient for overland flow
40	c_factor	real	0.0	none	cover and management factor scaling for overland-flow erosion
41	ch_d50	real	0.0	mm	median particle diameter of the main channel
42	co2	real	400.0	ppm	CO2 concentration at start of simulation
43	day_lag_mx	int	0	days	maximum days to lag hydrographs for HRU, RU, and channels on non-draining soils
44	igen	int	5	none	random generator code. `0` use defaults, `1` generate new numbers every run

Example

refdata/Ames_sub1/parameters.bsn:

```
parameters.bsn: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
  lai_noevap       sw_init      surq_lag      adj_pkrt  adj_pkrt_sed       lin_sed       exp_sed      orgn
     3.00000       0.50000       4.00000       1.00000       1.00000       0.00000       1.00000       0.0
```

Notes:

- Tool header names (`lai_noevap`, `sw_init`, `surq_lag`, `adj_pkrt`, `orgn_min`, `n_uptake`, `evap_adj`, `denit_exp`, `denit_frac`, `man_bact`, `adj_uhyd`, `n_fix_max`, `rsd_decay`, `rsd_cover`, `uhyd_alpha`, `splash`, `rill`, `surq_exp`, `cov_mgt`, `cha_d50`, `day_lag_max`) do not match the field names in the source (`evlai`, `ffcb`, `surlag`, `adj_pkr`, `cmn`, `n_updis`, `evrch`, `cdn`, `sdnco`, `bact_swf`, `tb_adj`, `nfixmx`, `decr_min`, `rsd_covco`, `uhalpha`, `eros_spl`, `rill_mult`, `eros_expo`, `c_factor`, `ch_d50`, `day_lag_mx`). The reader uses position only.
- Position 8 in this dataset is `orgn_min = 0.0`. In the source this is `cmn`, with default `0.003`. The example overrides the default to zero, which disables active organic-N mineralization.
- Position 7 is `exp_sed = 1.0`. This is `spexp` in the source, marked "not used".
- Positions 28 to 30 (`dorm_hr`, `plaps`, `tlaps`) are all zero in this dataset. The default for `tlaps` is `6.5` deg C/km; this dataset disables temperature lapse.

Related files

- `codes.bsn`. Method switches that select which of these parameters are active.